



TMscNet: a model with multiple information interaction for COVID-19 X-ray classification

Xicheng Wang¹ · Shuai Zhang¹ · Dejian Wei¹ · Junzhong Zhang² · Hui Cao¹

Received: 6 May 2024 / Accepted: 26 April 2025 / Published online: 5 June 2025

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Abstract

In recent years, large-kernel Convolutional Neural Networks (CNNs) have received significant attention in the classification of COVID-19 X-ray images. However, there are still unresolved issues. Existing large-kernel convolutional neural networks tend to overly emphasize local perceptual abilities, while the question of whether these networks possess strong general perceptual ability at multiple scales in the visual domain requires further exploration. Additionally, current neural network architecture designs primarily focus on spatial feature extraction from static images and have limited capability to understand temporal dynamic information in image sequences. To overcome these challenges, we introduce TMscNet, a tailored deep learning model for COVID-19 X-ray image classification. TMscNet introduces a new convolutional structure called DiverseConvBlock on top of the ConvNeXt network. It employs a novel self-tuning algorithm to fuse feature maps generated from convolutions with different kernel sizes. Furthermore, we employ Global Response Normalization to concurrently capture contextual and global image features, thus enhancing scale-based feature perception and extraction efficiency. Additionally, we introduce the Temporal Convolutional Network (TCN) to enable better capture and utilization of dynamic information in image sequences by leveraging its efficiency in handling long-term dependencies. Our model achieves high accuracy, specificity, and sensitivity on the publicly accessible Chest X-Ray Dataset and COVID-19 Radiography Database. In the classification of the first dataset, our model demonstrated an accuracy of 98.14%, a specificity of 98.79%, and a sensitivity of 98.29%. Upon validating with the second dataset, it achieved an accuracy of 98.05%, with specificity and sensitivity reaching 99.01% and 98.12%, respectively. These impressive results confirm the outstanding performance of our proposed network in COVID-19 and pneumonia image classification tasks.

Keywords COVID-19 X-ray image · Deep Learning · Global Information · Multiple Scales · Temporal Information

1 Introduction

COVID-19 is a highly contagious disease caused by the severe acute respiratory syndrome coronavirus (SARS-CoV-2) [1]. The virus was first identified in December 2019 in Wuhan, China, from three patients with severe respiratory disease and pneumonia [2]. This major discovery immediately became the center of global attention and aroused deep concern within the international community about the

disease. To effectively combat COVID-19, governments and leaders worldwide have actively adopted a variety of unique strategies, policies, and lifestyles [3]. The rapid development and implementation of these new strategies have been significantly driven by advances in science and technology. Chest X-ray image datasets are commonly utilized for identifying chest diseases like pneumonia, tuberculosis, and other lung-related ailments. These datasets offer several advantages, including low cost, wide availability of X-ray equipment, non-invasiveness, short examination duration, and affordability. Given the ongoing global healthcare crisis, X-ray images are a useful tool for widespread, accessible and relatively quick,

diagnostic support, often used alongside other methods, such as PCR tests, in managing epidemics like COVID-19. In recent years, researchers have designed various deep learning diagnostic models for COVID-19 to detect novel

✉ Xicheng Wang
a253285497@gmail.com

¹ College of Intelligence and Information Engineering,
Shandong University of Traditional Chinese Medicine,
Jinan, China

² The First Clinical College of Shandong, University
of Traditional Chinese Medicine, Jinan, China

coronavirus pneumonia using advanced artificial intelligence methods. He et al. [4] previously employed Convolutional Neural Networks for image categorization in chest X-ray classification, and several studies have achieved success in the diagnosis of COVID-19 [5] and other diseases [6]. Mpesiana et al. [7] utilized various transfer learning methods to detect COVID-19 using X-ray images. The system employed a dataset consisting of 224 images of patients with COVID-19, 700 images of patients with bacterial pneumonia, and 504 images of normal subjects. By combining X-ray imaging with deep learning techniques, the system successfully extracted important biomarkers associated with COVID-19, achieving optimal accuracy, sensitivity, and specificity of 96.78%, 98.66%, and 96.46%, respectively. However, despite these advancements, current research on lung radiographic image classification still faces several challenges:

- a) Diagnostic accuracy still needs to be improved by reducing misclassification.
- b) Most deep learning models have excessively complex structures for sequence modeling, leading to increased computational and training complexity.
- c) Additionally, many deep learning models are computationally inefficient when handling large-scale data due to the use of smaller convolutional kernels, which may not efficiently capture complex relationships in the data. As a result, training and inference on large-scale datasets can be relatively slow, potentially impacting accuracy.

Current deep learning models face significant limitations in both diagnostic accuracy and computational efficiency when applied to lung disease imaging tasks, such as detecting COVID-19 and pneumonia. The primary challenges include reducing misclassification rates, efficiently modeling sequence information, and handling complex data relationships with computationally efficient methods. To address these challenges, this study introduces TMscNet, a novel diagnostic framework designed to enhance classification performance. TMscNet effectively integrates both global and local feature information, capitalizing on the synergy between broad contextual understanding and the capture of fine-grained local details. The model uses a parallel design of large and small convolutional kernels to optimize computational efficiency while maintaining adaptability to varying data characteristics. Furthermore, it incorporates techniques focused on sequence information, enabling the model to capture intricate relationships within Chest X-ray Datasets. The main contributions of this study are summarized as follows:

- We introduce the GRN structure to enhance the model's ability to perceive contextual information and global features, improving overall image analysis.
- We propose the DiverseConvBlock(DCBlock), which optimizes feature extraction by combining convolutional kernels of different sizes, allowing the model to better adapt to various data characteristics.
- We enhance inter-channel communication and spatial aggregation by incorporating the Squeeze-and-Excitation (SE) attention module, which improves the model's performance by focusing on critical features.
- We integrate the TCN network structure to strengthen the models capability to process dynamic information in image sequences, achieving better performance in both training and inference.

The rest of this study is structured as follows: Sect. 2 will provide an overview of the related work, Sect. 3 will offer an in-depth discussion of the proposed methodology, Sect. 4 will present the results of the experiments and analyze them in detail, and finally, Sect. 5 will summarize the conclusions of the proposed work and outline future directions.

2 Related work

In the research area of utilizing deep learning for COVID-19 detection, Rahimzadeh et al. [8] developed a tandem CNN using the Xception and ResNet50V2 models for the classification of COVID-19 cases based on chest X-ray images. The dataset included 180 images of patients with COVID-19, 6054 images of patients with pneumonia, and 8851 images of normal subjects. Experimental results demonstrate that the system achieved an accuracy of 99.56% and a recall of 80.53%. Kumar et al. [9] introduced a deep learning approach dedicated to classifying COVID-19-infected patients by analyzing chest radiographs. The method utilized nine pre-trained models for feature extraction and combined them with SVM for final classification. The dataset consisted of 158 X-ray images each for COVID-19 and non-COVID-19 patients. The synergistic model of ResNet50 and SVM demonstrated excellent performance, statistically outperforming other models. Irfan et al. [10] proposed a hybrid deep neural network (HDNN) comprising dropout, convolutional, maximum pooling, Long Short-Term Memory (LSTM), and fully connected layers. They conducted experiments on a multimodal dataset consisting of 1700 normal images (1100 X-ray images and 700 CT images), 2500 COVID-19 images (1900 X-ray images and 600 CT images), and 2000 X-ray images of pneumonia. The study's results demonstrate that this network delivers more accurate and reliable results, achieving an accuracy rate of 99%. Alshazly et al. [11] explored the effectiveness

of deep learning models on chest CT images for rapid and automated COVID-19 diagnosis. They employed advanced network architectures and a transfer learning strategy with custom-sized inputs tailored for each architecture. Their models demonstrated superior performance on the SARS-CoV-2 CT-scan and COVID19-CT datasets, achieving high accuracy and other metrics. The models also provided accurate localization of COVID-19 regions, as validated by expert radiologists.

In the research area of COVID-19 diagnosis based on temporal models, Islam et al. [12] employed a combined deep CNN-LSTM network approach for the automated diagnosis of chest X-ray images of new cases of COVID-19 pneumonia. The CNN extracted complex features from the images, while the LSTM acted as a classifier. The system achieved 99.4% accuracy and 99.2% specificity on a dataset of 1525 images, including COVID-19, pneumonia, and normal images. El-Dahshan et al. [13] opted for a simpler and more straightforward TCN structure compared to LSTM networks. The model cleverly integrates the Empirical Wavelet Transform (EWT), ResNet50, and TCN. This combination maximizes the benefits of shared convolutional kernels and parallel computation, effectively reducing computation time and achieving 99.5% accuracy for COVID-19 in a triple classification task. Waseem et al. [14] employed a neural network and TCN to accurately identify different variants of COVID-19. The model initially encodes the sequences using numerical descriptors, followed by discriminative feature extraction of the encoded sequences using convolutional operations. TCN gathers sequential relationships between the features to enhance the accuracy of COVID-19 variant classification. This approach achieved 89.2% accuracy in the proposed task. In this paper, we propose a deep learning model that integrates temporal, local multi-scale, and global information to enhance the accuracy of predicting COVID-19-related X-ray images. Hamza et al. [15] developed a CNN-LSTM framework for COVID-19 diagnosis, addressing multisource fusion and redundant features. They combined Modified EfficientNet B0 with CNN-LSTM and used an improved max value optimization method. Their approach achieved higher accuracy by enhancing image contrast and dataset augmentation, demonstrating better performance on publicly available datasets compared to existing methods.

In the area of recent advancements in deep learning technology, Wang et al. [16] proposed the Deep Nearest Centroid Classifier (DNC). This network utilizes sub-centroids of training samples to describe class distributions and performs classification in the feature space based on the proximity between test data and class sub-centroids. DNC demonstrates excellent performance in image classification and pixel-level recognition tasks while also offering high transparency and interpretability. Additionally, Wang et al.

[17] designed a new training framework that enhances the performance of query-based instance segmentation models by learning discriminative query embeddings. This method has significantly improved the performance of several mainstream models (such as CondInst, SOLOv2, etc.) on the COCO and LVISv1 datasets. In the field of video instance segmentation, Liu et al. [18] introduced the Single-Stage Spatial Granularity Network (SG-Net). This network significantly improves the performance of detection, segmentation, and tracking tasks through dynamic sub-region mask prediction and target centrality motion modeling, achieving outstanding results on the YouTube-VIS dataset.

TMscNet represents a significant advancement in the field by integrating temporal, local multi-scale, and global information to enhance COVID-19 diagnosis through X-ray images. Unlike existing methods that focus on single aspects of feature extraction or temporal modeling, TMscNet combines these elements to achieve superior performance. It effectively captures long-term dependencies and global context through its unique combination of TCN and feature extraction techniques, outperforming previous models in accuracy, sensitivity, and robustness. This integrated approach allows TMscNet to handle diverse lesion types and dataset characteristics with exceptional generalization capability, setting a new standard in the field of COVID-19 detection.

3 Methodology

Figure 1 illustrates the system for COVID-19 detection, which consists of several stages. First, the original lung X-ray images undergo preprocessing, including data resizing, random rotation, and normalization. Additionally, data augmentation techniques, such as CLAHE enhancement, color jittering, and random horizontal flipping, are applied to address class imbalance and improve model generalization. CLAHE enhances the local contrast of the images, making important details more visible, especially in areas with low contrast. Color jittering introduces variations in brightness, contrast, and hue to simulate diverse imaging conditions, while random horizontal flipping increases the diversity of the training set by mirroring the images.

Afterwards, the preprocessed dataset is divided into a training set and a test set. The training data is then utilized to train the TMscNet architecture. This architecture categorizes the images into three classes: COVID-19, pneumonia, and normal. During the training process, the cross-entropy loss function is utilized to measure the disparity between the model output and the true labels. The model parameters were then updated using the AdamW optimizer. At the end of each training cycle, the training accuracy and loss are

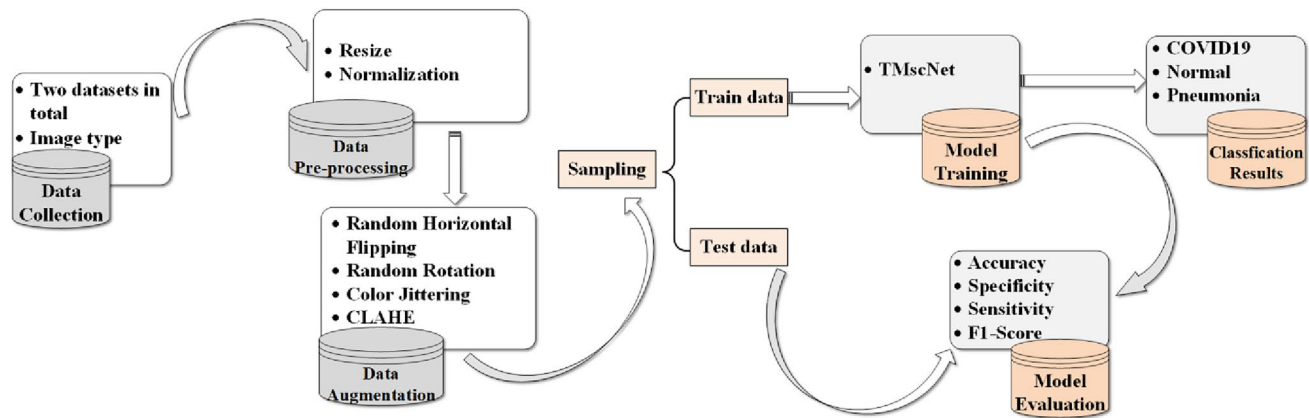


Fig. 1 An overview of the architecture of the proposed novel coronavirus detection system

Table 1 The division of the Chest X-ray Dataset

Dataset 1	COVID-19	Normal	Pneumonia	Total
Train	460	1266	3418	5144
Validation	116	317	855	1288
Total	576	1583	4273	6432

Table 2 The division of the COVID-19 Radiography Database

Dataset 2	COVID-19	Normal	Viral Pneumonia	Total
Train	900	1005	1008	2913
Validation	300	336	337	973
Total	1200	1341	1345	3886

evaluated, while validation accuracy and loss are calculated using the validation set. The performance of the proposed system is evaluated using metrics such as accuracy, specificity, sensitivity, and F1-score. In addition to the mentioned evaluation methods, we utilized the confusion matrix, Score-weighted Class Activation Mapping (Score-CAM) [19] technique, and ROC images to further evaluate system performance. The confusion matrix offers an intuitive comparison of classification results, while the Score-CAM technique visualizes the model's focal region for image classification. Meanwhile, ROC images provide insight into the model's performance across various thresholds.

3.1 Dataset Description

The first dataset used in this study is from Kaggle [20]. It comprises a total of 6,432 X-rays in JPEG format, including 4,273 pneumonia images, 1,583 normal images, and 576 COVID-19 images. The data were divided into 80% for training and 20% for testing, as detailed in Table 1.

The second dataset is sourced from an open-access chest X-ray image database on Kaggle [21]. It comprises three categories: COVID, viral pneumonia, and normal. The dataset includes 1,200 COVID-19 positive images, 1,345 viral pneumonia images, and 1,341 normal images. In this experimental setup, 75% of the data was designated for training while the remaining 25% was used for testing, as detailed in Table 2. Representative sample images from the three categories across both datasets are illustrated in Fig. 2.

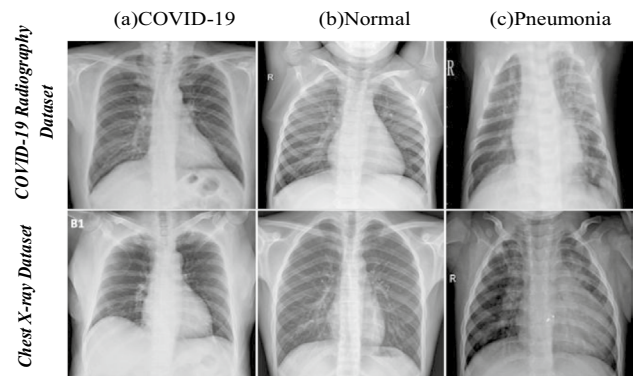


Fig. 2 Classify corresponding images from the two datasets into three columns: **a** COVID-19, **b** Normal, and **c** Pneumonia

3.2 Framework of TMscNet

TMscNet utilizes ConvNeXt [22] as its backbone network, incorporating the strengths of various Transformer structures through adjustments in network structure settings and parameter selection. Optimizations in training parameter settings, optimizer, and convolutional kernel size have resulted in superior performance compared to Swin Transformer (a Transformer variant combining self-attention with convolution) [23] on the ImageNet-1 K dataset. The backbone structure comprises four stages, each with a ratio of 3:3:9:3. The number of channels in each stage follows the sequence of

96, 192, 384, and 768, with corresponding feature map sizes of 56×56 , 28×28 , 14×14 , and 7×7 . Retaining the inverse bottleneck layer structure, where the middle layer is larger and the two ends are smaller, aids in effective avoidance of information loss. The framework structure of ConvNeXt is presented in Fig. 3.

TMscNet utilizes 2D convolution and layer normalization for data preprocessing, with its main network composed of multiple modules. TMscNet integrates DCBlock, which employs a parallel combination of large-kernel and small-kernel convolutions, replacing the traditional ConvNeXt convolutional layers. DCBlock uses large-kernel convolutions to capture broader contextual information while small-kernel convolutions focus on extracting detailed features. Fig. 4 illustrates the configuration of DCBlock with a large kernel size $K=9$. In the overall structure of TMscNet, the GRN structure and linear transformation further refine global feature extraction, which is then fused with the local multi-scale features extracted by DCBlock. This fusion strategy enhances the integration of both global and local information, thereby improving the model's ability to represent image features. Simultaneously introducing SEBlock [24] avoids structural complexity while achieving inter-channel communication and spatial aggregation, enhancing the network's adaptability to inter-channel dependencies. For a given input x , the output $S(\cdot)$ of SEBlock is represented by Eq. (1).

$$S(x) = \sigma(W_2 \delta(W_1 x)) \quad (1)$$

where σ is the sigmoid activation function, W_1 is the weight matrix of the first fully connected layer, δ is the Rectified Linear Unit (ReLU) [25] activation function. W_2 is the weight matrix of the second fully connected layer.

Downsample operations are applied at the beginning of the 2nd, 3rd, and 4th stages. The Downsample structure consists of Layer Normalization (LN) and a convolutional layer with a kernel size of 2×2 and a stride of 2. The Downsample operation first applies LN to the output of the previous stage for normalization, ensuring feature consistency and improving training stability. Then, the spatial dimensions of the feature map are reduced using the convolutional layer, which not only reduces computational

cost but also effectively preserves key features, helping the model focus on higher-level abstract representations. The Gaussian Error Linear Unit (GELU) [26] is introduced as an activation function between the GRN and linear transformation layers. This arrangement maximizes the utilization of the GELU activation function's output characteristics and complements the GRN structure, thereby enhancing the model's generalization ability and learning effectiveness. The mathematical expression of GELU is shown in Eqs. (2) and (3).

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (2)$$

$$\text{GELU}(x) = x \cdot \varphi(x) = \frac{x}{2} \cdot \left(1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right) \quad (3)$$

where $\text{erf}(x)$ represents the error function and $\varphi(x)$ denotes the cumulative distribution function of the standard normal distribution. The GELU function introduces nonlinear smoothing based on approximating ReLU, enhancing the model's expressive capability, particularly in handling large datasets. Its nonlinear properties are crucial for improving model performance.

A TCN block is introduced at the end of the second stage to operate at a lower feature map resolution. This enables better capture and utilization of dynamic information in the image sequence, aiding the model in understanding and abstracting high-level features. Subsequently, a global average pooling layer aggregates the features and reduces dimensionality. Further feature abstraction is achieved through a LN layer and a second TCN block. Finally, classification results are obtained via a softmax layer.

$$\text{Softmax}(x)_i = \frac{e^{x_i}}{\sum_{j=1}^C e^{x_j}} \quad (4)$$

In the formula for the softmax activation function, x_i represents the input score for category i , e^{x_i} is the exponential of the score for category i , C is the total number of categories, and j is the index used for calculating the sum of exponentials of the scores across

Fig. 3 ConvNeXt Framework Architecture Diagram

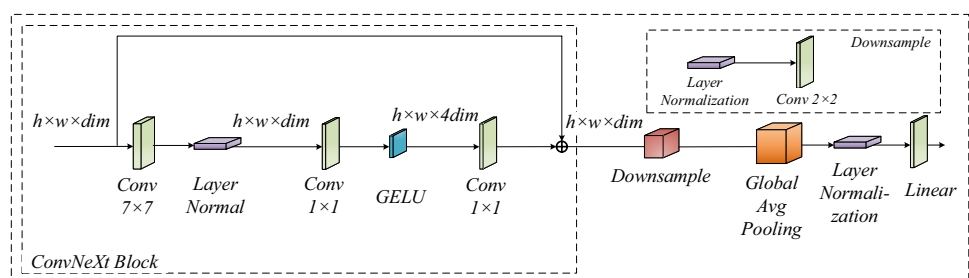
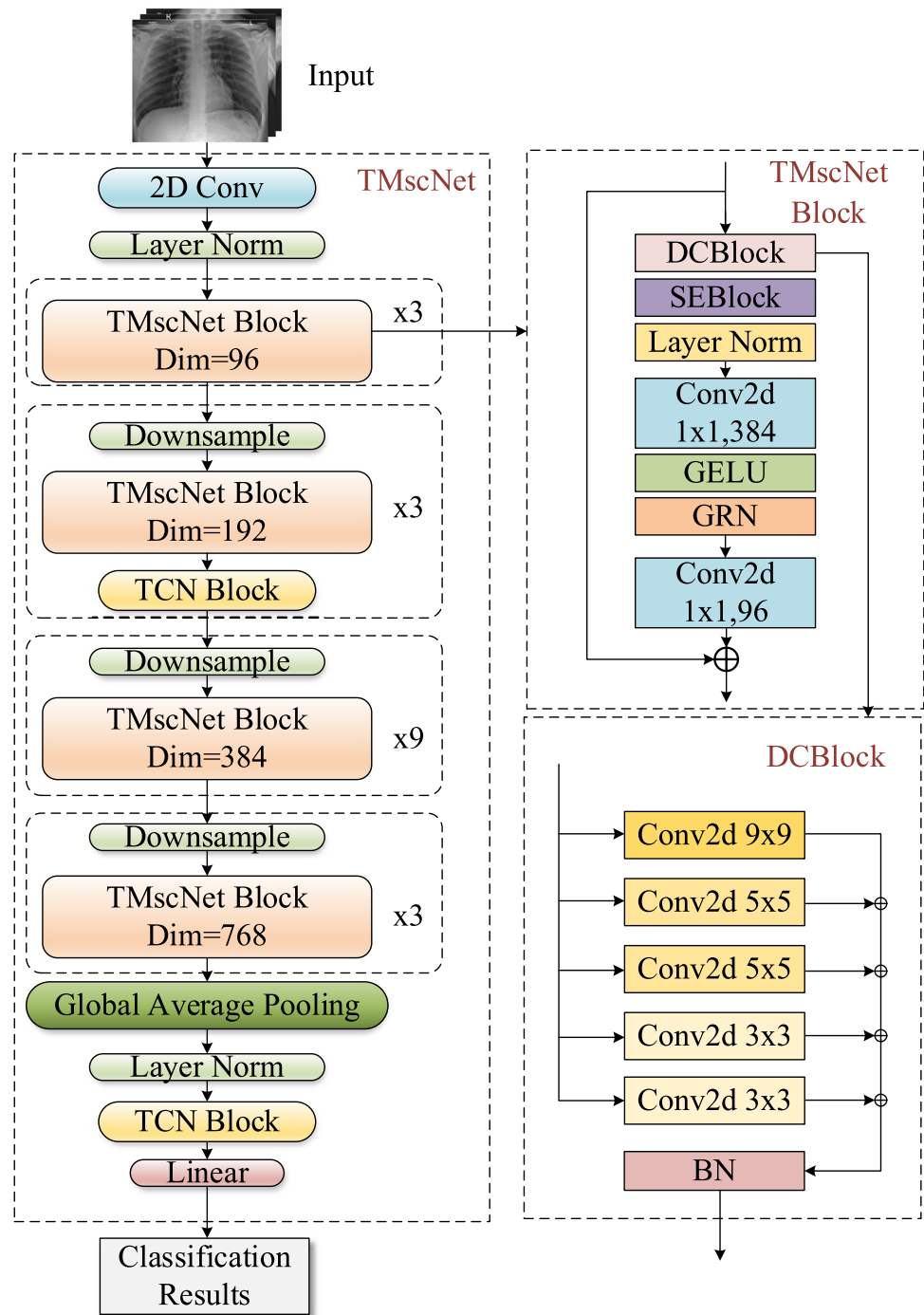


Fig. 4 TMscNet Framework Architecture Diagram



all categories. The formula computes the predicted probability for each class by normalizing the exponential of the input scores with the sum of the exponentials of all input scores, ensuring that the probabilities for all classes sum to 1.

3.3 Global response normalization

In this section, we introduce a new technique called Global Response Normalization (GRN) [27] to enhance the effectiveness of combining the DCBlock with the ConvNeXt architecture. With the GRN structure, we can better capture the global patterns and correlations in the dataset, thereby enhancing the model's generalization ability and performance.

Regarding the GRN implementation, for a given input feature $X \in R^{H \times W \times C}$, the GRN layer consists of three steps: global feature aggregation, feature normalization, and feature calibration. The GRN unit utilizes the computed feature normalization scores to calibrate the raw input response, as implemented below:

$$\zeta(X) = X \in R^{H \times W \times C} \rightarrow gx \in R^C \quad (5)$$

First, a spatial feature map X_i is aggregated into a vector gx using a global function $\zeta(X)$. This step can be likened to a simple pooling layer. Norm-based feature aggregation, specifically using L2-norm, yielded the best performance in experiments. This aggregation process results in a set of aggregated values gx :

$$gx = \zeta(X)_i = \{\|X_1\|, \|X_2\|, \dots, \|X_C\|\} \in R^C \quad (6)$$

Here, gx is a vector containing these scalars. Each $\|X_i\|$ represents a scalar that aggregates the statistics of the i -th channel, summarizing the feature information of that channel. Next, we apply the response normalization function $N(\cdot)$ to the aggregated values using standard deviation normalization, as follows:

$$N(\|X_i\|) = \|X_i\| \in R \rightarrow \frac{\|X_i\|}{\sum_{j=1, \dots, C} \|X_j\|} \in R \quad (7)$$

The above X_i represents the L2 norm for channel i . This step computes the relative importance of each channel compared to others, inducing a feature competition across channels. Research indicated that simple split normalization yielded the best results. Finally, the computed feature normalization scores were utilized to calibrate the raw input response.

$$X_i = X_i * N(\zeta(X)_i) \in R^{H \times W} \quad (8)$$

To simplify the optimization process, two additional learnable parameters, γ and β , are introduced with initial values set to zero. In the context of our network model, these parameters are applied within the GRN layer to enhance its adaptability during training. Specifically, residual connections are incorporated between the inputs and outputs of the GRN layer to facilitate smoother optimization. Consequently, the final structure of the GRN block is as follows, allowing the GRN layer to initially perform constant functions and gradually adapt during the training process. The final result is shown as Eq. (9). The pseudocode of the GRN structure is shown in Textbox 1.

$$X_{output} = \gamma * X * N(\zeta(X)) + \beta + X \quad (9)$$

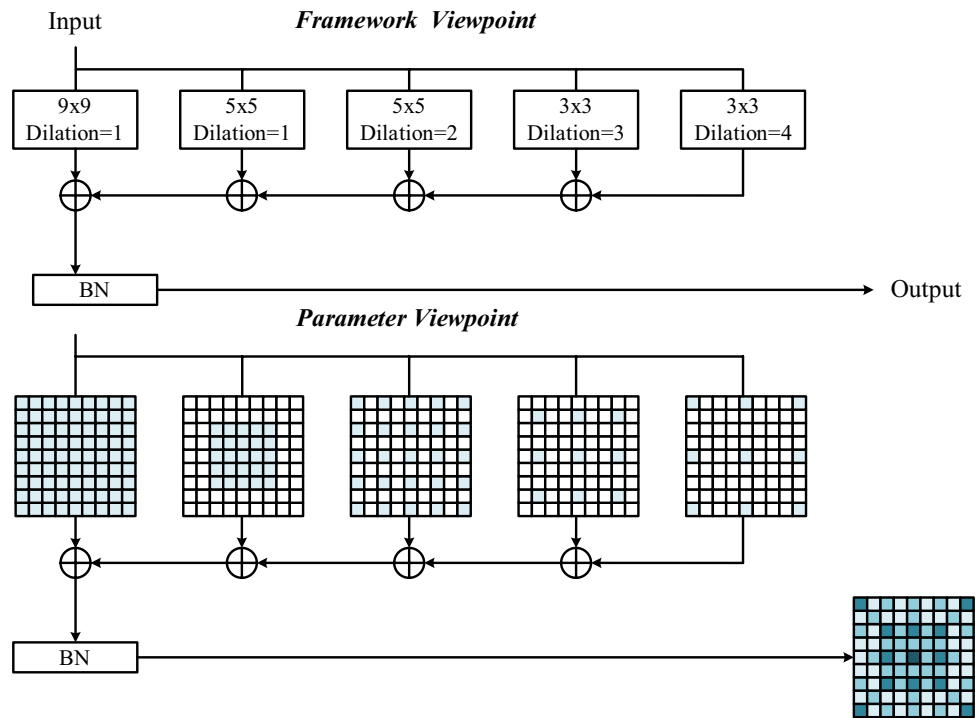
3.4 DiverseConvBlock

In this section, we introduce a new convolutional structure called DiverseConvBlock (DCBlock). DCBlock replaces the convolutional layers in the original ConvNeXt model, aiming to improve the efficiency of the feature extraction process. In early convolutional neural networks, large kernel convolutions were widely employed in classical ConvNet architectures such as AlexNet [28] and Inception [29], often utilizing large 7×7 or 11×11 kernels in the lower layers. However, the use of large kernels declined with the advent of subsequent models such as VGG-Net [30]. Nowadays, very large kernel convolutional neural networks have become popular, mainly focusing on making large kernels even bigger [31] and applying them to areas such as multi-task learning. Nevertheless, as the size of the convolutional kernel increases, the number of parameters and the computational complexity of the network also increase dramatically. This may lead to the training and inference process of the model becoming more time-consuming and resource-intensive. Additionally, the fixed size of the large kernel may restrict the network's ability to extract features at various scales. This limitation, in turn, can impact the network's performance in handling complex scenes.

Research has shown that large kernel convolution should be used in conjunction with parallel small kernel convolution to capture small-scale patterns effectively during training [32]. The outputs of these convolutions are summed before passing through a Batch Normalization (BN) layer [33]. Subsequently, after training, the BN layer is merged into the convolutional layer using the Structural Re-parameterization method [34]. This merging ensures equivalent incorporation of the small kernel convolution into the large kernel convolution for inference. To alleviate the computational burden of the additional dilated convolutional layer during inference, we consolidate the entire block into a single convolutional layer without dilation. This consolidation entails treating the omitted input pixels as additional zero values inserted into the convolutional kernel. Consequently, a dilated convolutional layer with a small kernel can be transformed equivalently into a non-dilated layer with a sparsely larger kernel (i.e., $r = 1$). Let the kernel size of the dilated layer be k . By inserting zero values, the kernel size of the corresponding non-dilated layer becomes $(k - 1)r + 1$, known as the equivalent kernel size [35]. The conversion of the original kernel W with size $\mathbf{k} \times \mathbf{k}$ into a new kernel W' with size $(k - 1)r + 1$ can be achieved by employing a transposed convolution with a stride of r and a unit kernel $\mathbf{I} \in \mathbf{R}^{1 \times 1}$. This process can be illustrated in PyTorch-style pseudo-code as follows:

$$W_i' = \text{conv.transpose2d}(W_i, k_i, r_i) \quad (10)$$

Fig. 5 DCBlock structure diagram, in this example we take $K=9$ as an exemplar, but we can also use larger values of K to build further extension layers



Utilizing this equivalent transformation, we implement an extended DiverseConvBlock. This block is based on non-expanded small kernels and incorporates multiple expanded small kernel layers to complement a non-expanded large kernel transformation layer. The hyperparameters of this block include the size K of the large kernel, the kernel size k of the multiscale convolutional layers, and the dilation rates r . Figure 5 illustrates the internal structure of the DiverseConvBlock from two perspectives: the Framework Viewpoint and the Parameter Viewpoint. The Framework Viewpoint shows the overall architecture of the block, which utilizes a main convolutional kernel $K=9$ in combination with several smaller kernels. These kernels are integrated using different dilation rates $r=(1,2,3,4)$ and kernel sizes $k=(5,5,3,3)$ to capture features at multiple scales. On the other hand, the Parameter Viewpoint delves into how the smaller convolutional kernels are expanded using dilated convolutions, forming a larger effective kernel. The figure demonstrates the transformation of the kernels under varying dilation rates and illustrates how the dilated kernels are summed together to create the final combined convolution effect.

For larger values of k , we can increase the kernel size and dilation rate in the expansion layer and can also increase the number of expansion layers. The selection of kernel size and dilation rate is flexible, but they are bounded by $(k-1)r+1 \leq K$. In this experiment, five multiscale convolutional layers are utilized with parameters set to $K=13$, $r=(1,2,3,4,5)$, and $k=(5,7,3,3,3)$. Their equivalent kernel sizes are $(5,13,7,9,11)$. To convert the DiverseConvBlock into a large kernel transformation layer

for inference, we first transform each extension layer into a function, and then sum up all the resulting kernels with appropriate zero padding.

$$\text{Combined}(x) = \sum_{i=1}^n \text{ZeroPadding}((W_i)) \quad (11)$$

W_i is zero-padded using the ZeroPadding operation to ensure that the features at each scale can be fully extracted. $\text{Combined}(x)$ is obtained by summing the obtained equivalent kernel W_i . Meanwhile, in order to accommodate feature extraction at different scales, the overall input is subsequently fed into a BN layer. For the input x , the output of DCBlock can be expressed by Eq. (12).

$$\text{Output} = \vartheta(\text{BN}(\lambda(x) + \text{Combined}(x))) \quad (12)$$

In the formula, $\vartheta(\cdot)$ denotes the Structural Re-parameterization operation, and $\lambda(\cdot)$ denotes a convolution with kernel size K . As an example, in Fig. 5, a layer with $k=3$ and $r=2$ is converted to a sparse 7×7 kernel, which is then added to a 9×9 kernel with one pixel of zero padding on each side. For a layer with $k=5$, $r=1$, there are two pixels of zero padding on each side.

3.5 Temporal convolutional network

Recurrent Neural Networks (RNNs) have become the method of choice for sequence modeling tasks due to their superior ability to capture temporal relationships. RNN variants such as LSTM [36], Convolutional LSTM (ConvLSTM) [37], WaveNet [38], and Gated Recurrent Units (GRU) [39] play

important roles, effectively capturing long-term dependencies and achieving leading-edge results in sequence modeling tasks. The TCN structure [40] improves upon the CNN model by adapting it for sequence modeling tasks and incorporating causal constraints. TCN outperforms models like RNN, LSTM, and GRU due to its simpler and more straightforward structure. Although TCNs are traditionally used for sequential data, they can be effectively applied in image classification tasks due to their ability to capture hierarchical spatial features through deep convolutional layers. TCNs utilize dilated convolutions, which allow the network to have a larger receptive field without increasing the number of parameters. This enables TCNs to detect intricate patterns and relationships within the spatial structure of images, such as textures and edges, which are critical for accurate classification. By leveraging the temporal dimension implicitly through the network's depth and dilation, TCNs enhance the model's ability to discern complex spatial hierarchies, leading to improved classification performance in tasks like COVID-19 X-ray image analysis. The simple and direct structure of TCN, along with its ability to handle sequences of different lengths, offers greater flexibility and scalability for practical applications. It also adopts the architecture of the 1D Fully Convolutional Network (FCN) [41], enabling it to process input sequences of varying lengths and produce output sequences of the same length. To maintain temporal order consistency, TCN utilizes causal convolution, where the output at time t is only influenced by input elements at time t and preceding instances. This process can be succinctly summarized as:

$$TCN = 1D FCN + \text{Causal Convolutions} \quad (13)$$

In the design of TCN, residual blocks are employed to mitigate performance degradation with increasing layer depth.

Each residual block comprises two layers of extended causal convolution and a nonlinear activation function, ensuring the network's effectiveness and stability. Additionally, ReLU is selected as the activation function, enhancing the network's nonlinear properties. Given an input x , the ReLU activation function produces an output of 0 when the input is less than 0 and an output equal to the input when the input is greater than 0. This method can be mathematically represented as shown in Eq. (14). The residual blocks in the TCN structure are presented as shown in Eq. (15).

$$\text{ReLU}(x) = \max(0, x) \quad (14)$$

$$H(l) = \text{ReLU}(F(H(l-1)) + (H(l-1))) \quad (15)$$

At this juncture, $H(l)$ represents the output of layer l , and $F(\cdot)$ denotes the convolution operation. The new output is obtained by summing the original output with the result of the convolution operation and then applying the ReLU activation function. In the design of TCN, 1×1 convolutions are utilized to address potential discrepancies between input and output widths. The typical architecture of TCN is illustrated in Fig. 6

The introduction of dilated convolution is important for network design. It adds weight to the convolution kernel, effectively expanding the range of time series viewed by the network while keeping the input data intact. In a given one-dimensional sequence with input $x \in R^n$ and filter $f: \{0, 1, \dots, k-1\} \rightarrow R$, the dilated convolution operation $F(\cdot)$ for sequence element t is defined as:

$$F(t) = (x *_d f)(t) = \sum_{i=0}^{k-1} f(i) \cdot x_{t-d \cdot i} \quad (16)$$

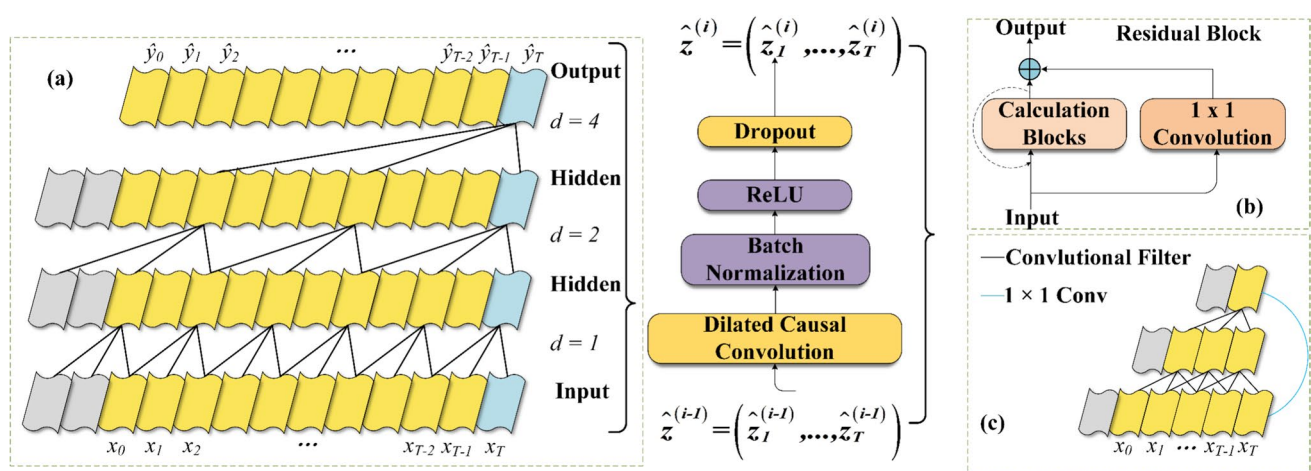


Fig. 6 Architectural elements in TCN. In **a**, we see dilated causal convolution with dilation coefficients $d=1, 2, 4$ and filter size $k=3$. This convolution operates with a receptive field that covers all values of the input sequence. Moving to **b**, we observe the TCN residual

block, where the dashed line denotes the reuse of calculation blocks. Lastly, **c** presents an example of residual connectivity in TCN ($k=3, d=1$). In the residual function, the black line represents the filter, while the blue line indicates the 1×1 convolution operation

The filter size k is defined by f , where the dilation coefficient is d , and $\mathbf{t} - \mathbf{d} \cdot \mathbf{i}$ denotes the past direction. When $d = 1$, the dilation convolution reduces to a standard convolution. Increasing the dilation coefficient allows the output of the top layer to reflect a wider range of inputs, thus extending the perceptual range of the TCN. So the TCN can expand its receptive field by increasing the dilation coefficient d or choosing a larger filter size k . Fig. 6 illustrates the dilated 1D convolution corresponding to the dilation coefficients $d = 1, 2, 4$ and the filter size $k = 3$. TCN also utilizes weight normalization to ensure the normalization of convolutional filter weights, enhancing both model convergence speed and generalization ability. Additionally, after each extended convolution, a spatial dropout layer is incorporated for regularization, where the entire channel is zeroed out in each training step.

4 Results and analysis of the experiment

4.1 Experimental setting

The deep learning model presented in this paper is implemented in Python programming language and developed using the PyCharm IDE. PyTorch 1.11 library was selected as the primary framework. The experiments were conducted on a system equipped with an Intel(R) Core(TM) i7-13620H CPU running at 2.40 GHz and a NVIDIA GeForce RTX-4060 GPU.

To ensure comparability with other models on this dataset, and to maintain fairness and consistency in the experiments, the dataset was partitioned using the official predefined training and testing sets. Prior to data augmentation, the original lung X-ray images underwent preprocessing, which included resizing to a consistent size and normalization to standardize pixel values. To address class imbalance in the dataset, we performed data augmentation to increase the number of samples. Specifically, we applied random rotation, random horizontal flipping, and color jittering techniques. These augmentations expanded each category in Dataset 1–3500 images and each category in Dataset 2 to 2000 images. Additionally, CLAHE contrast enhancement was applied as part of the data augmentation process. Through iterative tuning, the optimal parameters for the CLAHE function were found to be ClipLimit=4.0 and TileGridSize=(4, 4), which achieved the best performance. Table 3 provides a detailed overview of our training configurations to ensure transparency in the experiments.

To assess the performance of this model, a combination of evaluation metrics is employed, including accuracy, specificity, sensitivity, F1-score, and receiver operating

Table 3 Detailed training configurations of models

Settings	Value
Expanded Dataset 1 Size	10,500
Expanded Dataset 2 Size	6000
Input Scale	224
Batch Size	32
Optimizer	AdamW
Learning Rate	4×10^{-3}
Learning Rate Schedule	Cosine
Warmup Epochs	5
Epochs	200
Drop Path Rate	0.2
Random Rotation	15°
Random Horizontal Flip	0.5
Color Jittering	0.2
CLAHE ClipLimit	4.0
CLAHE TileGridSize	(4,4)

characteristic curve (ROC). The definitions of these metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \quad (17)$$

$$Specificity = \frac{TN}{TN + FP} \quad (18)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (19)$$

$$F_1 - Score = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (20)$$

TP represents the number of true positives, FP represents the number of false positives, FN represents the number of false negatives, and TN represents the number of true negatives. Furthermore, the loss function employed is the cross-entropy loss function, utilized to quantify the disparity between the model's prediction and the true label. The formula for the cross-entropy loss function is presented in Eq. (21).

$$CE Loss = - \sum_{c=1}^C y_c \log(\hat{y}_c) \quad (21)$$

where C is the number of classes, y_c is the one-hot encoded value of the true label; if the sample belongs to class c , then $y_c = 1$, otherwise $y_c = 0$, and $\hat{y}_c$ is the predicted probability of the model for class c .

4.2 Ablation experiment

4.2.1 Normalization structures

Experimental results have shown that various normalization techniques exhibit distinct impacts on model performance. In our study, we examined the impact of incorporating different normalization techniques into the TMscNet Block and assessed their performance across two datasets. However, we observed that traditional normalization techniques have certain limitations when confronted with complex datasets. For instance, they may not fully leverage global information and could significantly affect the model's generalization capability, consequently resulting in decreased accuracy. However, the introduction of GRN addresses the information gap encountered when the model handles complex datasets, underscoring the significance of integrating global information in model design. Experimental results indicate that GRN outperforms other normalization structures. Table 4 illustrates the accuracy results for the classification task with different normalization structures incorporated. Upon integrating the GRN structure, the model achieves accuracy rates of 96.20% and 96.12% on the two datasets, respectively.

4.2.2 Adjusting k sizes and SE structure in DCBlock

In our study of DCBlock, we initially examined the potential limitations of existing block structures, such as inadequate inter-channel communication and spatial aggregation, and performance deterioration when handling scale-specific input data. Upon exploration, we found that the SE Block

offers a more efficient implementation of inter-channel communication and spatial aggregation. Compared to alternative structures, the SE structure stands out for its efficient and straightforward design features. Its simplicity may help mitigate performance degradation caused by structural complexity, as observed in our experiments, where improved performance was noted. Based on these findings, we propose integrating the SE structure into our block design to enhance both the representation capability and performance of the model. In our experiments, the best results were achieved when using $k=13$ and incorporating the SE Block structure. Further details are presented in Table 5.

4.2.3 Evaluation of TCN at different stages

In reference to TCN networks, we initially introduced a TCN module to enhance the model's ability to process temporal information. However, experiments revealed limitations when placing the TCN solely between Fully Connected (FC) and LN layers, as it failed to effectively capture semantic information at various levels within the model. Therefore, we decided to incorporate TCN modules into various stages of the model. This decision was motivated by the aim to enable the model to encompass both low-level and high-level semantics, thereby enhancing its ability to capture dynamic information in image sequences more comprehensively. Introducing TCN modules at different stages allows us to

Table 4 The impact of different normalization structures on model performance

Accuracy(%)	BN	LRN	LN	GRN
Dataset 1	95.58	95.27	95.38	96.20
Dataset 2	95.81	95.73	95.49	96.12

Table 6 The performance of TCN is tested on the Chest X-ray Dataset at different stages with advanced semantic combinations. Here, Adv. S stands for Advanced Semantics, and S stands for Stage

TCN	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)
Adv.S	96.20	97.58	94.92	95.03
Adv.S + s1	96.89	97.86	97.08	97.04
Adv.S + s2	97.36	98.47	98.08	97.31
Adv.S + s3	96.27	97.98	97.22	96.29
Adv.S + s4	96.74	98.15	96.03	96.07

Table 5 The performance comparison of different kernel sizes and with and without the SE structure on the Chest X-ray Dataset was conducted. Kernel sizes of $k=5$, $k=7$, $k=9$, $k=11$, and $k=13$ were selected for comparison

Performance Evaluation	No SE				With SE				
	Kernel Size	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)
	5	96.04	97.92	96.99	98.91	96.51	97.90	95.89	95.53
	7	96.50	98.08	97.65	96.74	96.58	97.34	96.13	96.62
	9	96.35	97.92	96.94	96.44	96.35	97.41	95.60	96.09
	11	96.35	97.76	97.18	96.48	96.58	97.68	96.41	96.54
	13	96.66	97.94	96.97	96.64	97.20	98.22	97.37	97.06

explore their role in semantic representation more comprehensively and maximize their potential benefits.

Table 6 displays the test results of TCN in various contexts. After combining the two semantics, the accuracy of all four stages improves. Particularly, adding TCN after Stage 2 yields the most outstanding performance, with accuracy, specificity, sensitivity, and F1 scores reaching 97.36%, 98.47%, 98.08%, and 97.31%, respectively.

4.2.4 Validation of module effectiveness through ablation experiments

In order to further validate the effectiveness of each module proposed in this paper, we separately tested the performance of the base model and the combination of each structure, dividing the ablation experiments into five phases: a, b, c, d, and e. The results of the ablation experiments are summarized as follows. Specifically, in phase b, we individually added the global response normalization (GRN) structure to improve the global feature learning capability of the model. However, the results indicate that introducing global normalization alone cannot significantly improve the.

model performance, showing only a slight improvement compared to the original model. In stage c, we introduced DCBlock to interoperate with GRN. Here, the model

enhances local multi-scale feature extraction capability through multi-scale fusion convolution while conducting global feature normalization. This integration led to a 1.39 percentage point improvement in accuracy compared to stage a, further validating the effectiveness of combining DCBlock and GRN. Moving to stage d, by integrating GRN with TCN, we captured not only the global features in the spatial domain but also established "implicit sequential relationships" between different regions of the image. Although lung X-ray images are static, TCN can map image features into a one-dimensional sequence, allowing for the capture of dependencies between different regions. The experimental results showed that the accuracy in stage d reached 97.36%, validating the effectiveness of TCN in processing lung X-ray images.

The experimental results indicate that improvements made to the model in stages c and d have significantly enhanced the image classification results. Combining the.

models from stages c and d, we form our model TMscNet in stage e. This model is designed to encompass the abilities of local multiscale feature extraction, global feature normalization, and time series information processing. Although introducing these structures has increased the model's parameter count and computational complexity, these improvements have significantly enhanced the performance. Despite the increased complexity, the accuracy and

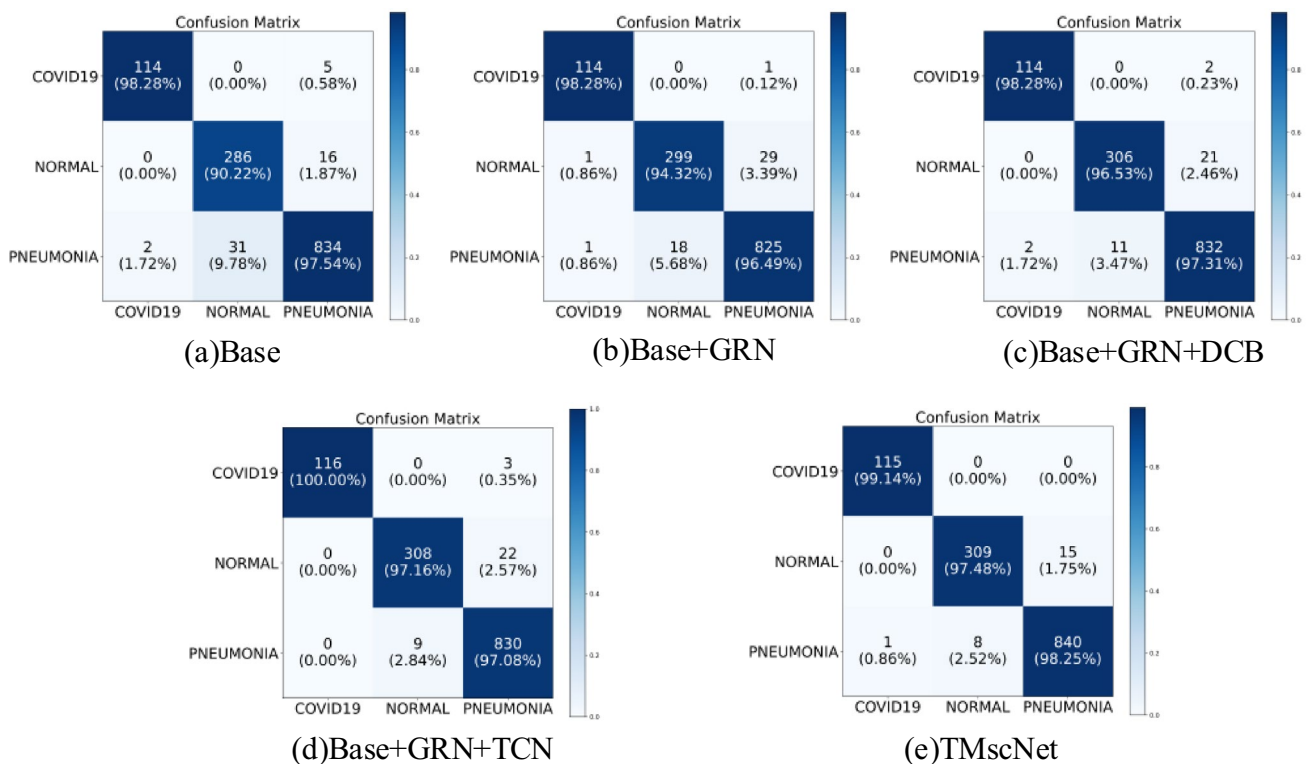


Fig. 7 Confusion matrix for the test set of ablation experiments on the Chest X-ray Dataset

Table 7 Model performance for ablation experiments

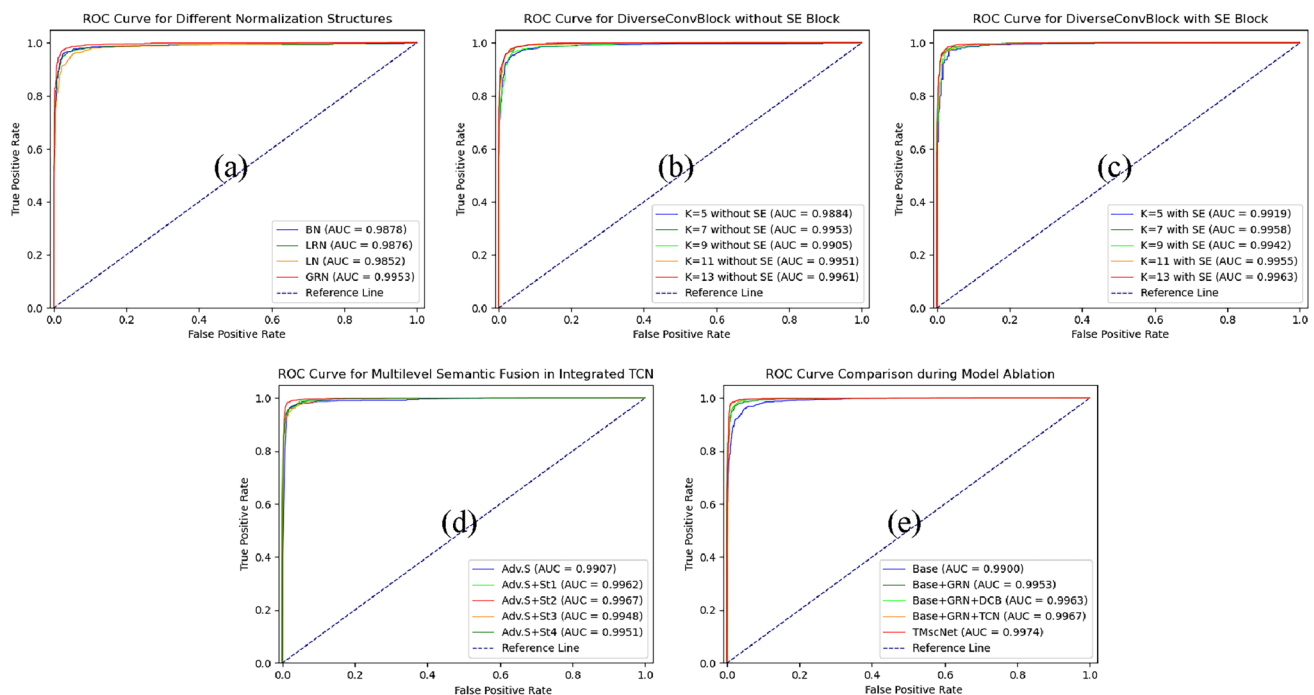
Stage	Model	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)	Δ Accuracy(%)	Param (M)	FLOPs(G)
(a)	Base	95.81	96.77	95.35	95.43		28.57	2.78
(b)	Base + GRN	96.12	97.48	96.36	96.13	$\uparrow$ 0.31	28.57	4.46
(c)	Base + GRN + DCB	97.20	98.22	97.37	97.06	$\uparrow$ 1.39	30.35	4.51
(d)	Base + GRN + TCN	97.36	98.47	98.08	97.31	$\uparrow$ 1.55	33.34	4.53
(e)	TMscNet	98.14	98.79	98.29	98.19	$\uparrow$ 2.88	35.35	4.66

other performance metrics have still shown notable improvements, demonstrating the effectiveness of the enhancements in boosting the model's capabilities. The accuracy, specificity, sensitivity, and F1 score of TMscNet reached 98.14%, 98.79%, 98.29%, and 98.19%, respectively. Compared with the base model, the accuracy increased by 2.88 percentage points, further enhancing lesion recognition performance in chest X-ray images. The confusion matrix for each stage of the ablation experiment on the test set is illustrated in Fig. 7 below. Additionally, Table 7 presents the result metrics of the ablation experiments conducted on the Chest X-ray Dataset, where Base refers to the ConvNeXt network and DCB represents our proposed DiverseConvBlock.

4.2.5 Receiver operating characteristic

The ROC curves, with True Positive Rate (TPR) as the vertical axis and False Positive Rate (FPR) as the horizontal

axis, demonstrate the classifier performance at different thresholds. In Fig. 8, each subplot serves a distinct purpose: (a) represents the performance comparison among different normalization structures, with the GRN structure exhibiting the best performance. Subplots (b) and (c) illustrate the performance of DCBlock with or without SE structure at different K values. Regardless of the presence of the SE structure, the performance for large K values surpasses that for small K values. Moreover, the overall Area Under Curve (AUC) in (c) slightly exceeds that in (b) after incorporating the SE structure. Notably, the performance of TCN in the Adv. Sem + st2 stage shown in (d) significantly outperforms other stages, highlighting the optimal results achieved by TCN with advanced semantics and stage 2 fusion. Finally, in (e), TMscNet demonstrates its superior performance with the introduction of structures such as global response normalization, local multi-scale information fusion, and temporal convolution, achieving an AUC of 99.74%.

**Fig. 8** ROC curves across various ablation experiment stages

4.2.6 Score-CAM

We employed the Score-CAM technique to investigate the visual interpretability of the proposed network. The results, depicted in Fig. 9, showcase visualizations emphasizing regions of significant interest to the model during the learning process. Figure 9's first column exhibits three classes of chest X-ray images from the Chest X-ray Dataset. Columns 2 to 4 illustrate the visualization outcomes of Base, Base + GRN, Base + GRN + TCN, and Base + GRN + DCB, respectively. Meanwhile, column 5 presents the visualization outcomes of TMscNet. Comparative analysis of the visualization outcomes from other models underscores TMscNet's pronounced advantage: its ability to precisely pinpoint lung lesion areas and capture details overlooked by other models. This highlights TMscNet's efficacy in identifying abnormal lung regions, thus enhancing classification accuracy. Score-CAM visualization results further affirm TMscNet's proficiency in identifying regions of interest, this enhances its ability to accurately categorize lung X-ray images.

Fig. 9 Score-CAM technology visualization results

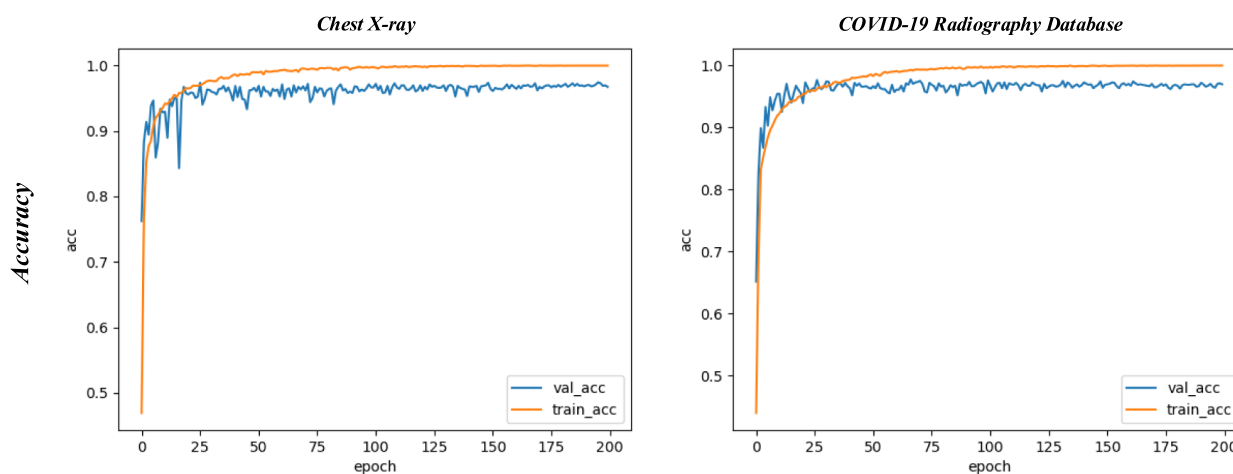
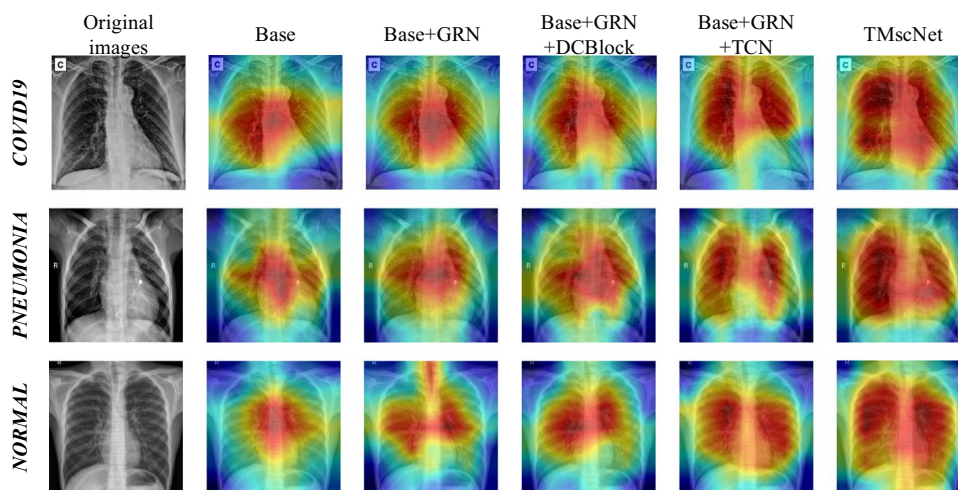


Fig. 10 Accuracy change curve of TMscNet on two datasets

Table 8 Demonstration of performance metrics for the three categories on the Dataset 1 test set

Type	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)
COVID-19	99.85	99.92	99.14	99.14
NORMAL	98.29	98.46	97.79	96.57
PNEUMONIA	98.14	98.15	98.13	98.59

4.3 Performance evaluation and comparison

To visualize the experimental results, we plotted the accuracy change curve of the TMscNet network on two datasets, as shown in Fig. 10. Observing the graph, we note a rapid increase in accuracy rate within the initial 50 rounds of training. Between 50 and 100 rounds, the curve gradually stabilizes, maintaining this stability until 200 rounds. Table 8 showcases the performance metrics of different classes on the COVID-19 Radiography Database dataset. Meanwhile, Table 9 presents a comparison between TMscNet and other mainstream algorithms. Incorporating structures such as global response normalization, multi-scale information fusion, and temporal convolution, it exhibits a significant advantage in model performance. Specifically, TMscNet achieved an accuracy of 98.14%, a specificity of 98.79%, a sensitivity of 98.29%, and an F1 score of 98.19%.

We conducted an in-depth study on the robustness of the TMscNet model using the COVID-19 Radiography Database. The Chest X-Ray Dataset covers a broader range of lung lesions, while the COVID-19 Radiography Database focuses more specifically on detecting COVID-19 infections. The experimental results demonstrate that, even with differences in image types and target classifications, the TMscNet model consistently exhibits remarkable stability across different categories, indicating its strong generalization capability in handling various lesion types and dataset characteristics. Table 10 showcases TMscNet's performance metrics on Dataset 2 across three categories. TMscNet achieves remarkable scores of 98.05% for Accuracy, 99.01% for Specificity, 98.12% for Sensitivity, and 98.10% for F1-Score. Its exceptional performance can be attributed to its unique approach, combining feature extraction and temporal order at different scales globally and locally, enabling superior

Table 9 Performance comparison between TMscNet and mainstream algorithms on Dataset 1

Model	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)	Params(M)	FLOPs(G)
MobileNet [48]	96.34	96.06	95.63	95.40	5.48	0.23
Abdar et al. [46]	96.35	96.54	96.37	96.36	–	–
VGG19 [48]	96.58	96.04	95.88	95.41	138.36	15.47
Jain et al. [47]	97.00	94.65	94.66	96.00	–	–
Lasker et al. [48]	97.28	97.68	97.65	96.97	–	–
RMT [44]	96.97	98.03	97.49	97.12	54.15	9.97
Biformer [45]	97.20	98.17	97.67	97.43	56.80	9.37
TMscNet(GRU)	96.20	97.70	96.24	95.72	24.30	3.54
TMscNet(LSTM)	96.51	97.96	96.84	96.32	24.70	3.54
TMscNet	98.14	98.79	98.29	98.19	35.35	4.66

Table 10 Demonstration of performance metrics for the three categories on the Dataset 2 test set

Type	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)
COVID-19	99.79	99.70	100.00	99.67
NORMAL	98.15	98.43	97.62	97.33
Viral Pneumonia	98.15	98.90	96.74	97.31

Table 11 Performance comparison between TMscNet and mainstream algorithms on Dataset 2

Model	Accuracy(%)	Specificity(%)	Sensitivity(%)	F1-Score(%)	Params(M)	FLOPs(G)
ResNet-101 [4]	93.01	93.04	93.01	93.02	44.55	7.87
DeiT-Base [42]	95.40	96.73	94.23	94.72	87.34	16.95
ConvNeXt [22]	95.81	96.77	95.35	95.43	28.57	2.78
ResMLP [43]	95.90	97.23	94.83	95.42	30.00	6.00
Swin Transformer [23]	96.30	98.13	96.35	96.37	28.29	4.37
Biformer [45]	97.01	98.48	97.07	97.09	56.80	9.37
RMT [44]	97.22	98.59	97.28	97.29	54.15	9.97
TMscNet(GRU)	96.09	98.02	96.12	96.12	24.30	3.54
TMscNet(LSTM)	96.30	98.12	96.33	96.38	24.70	3.54
TMscNet	98.05	99.01	98.12	98.10	35.35	4.66

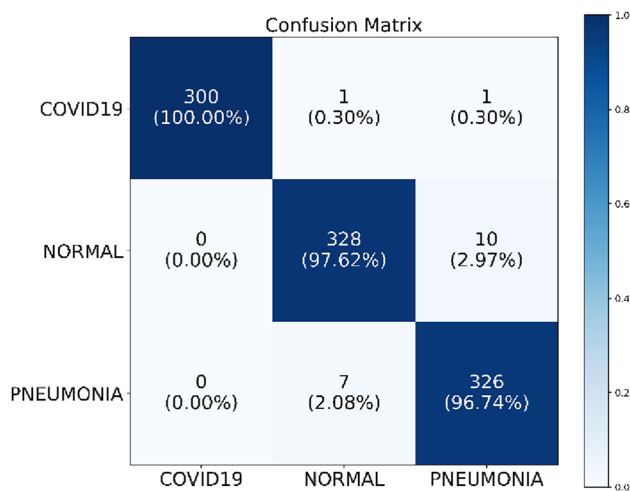


Fig. 11 Confusion Matrix for TMscNet on Dataset 2 Validation Set

recognition of lung lesion regions. Subsequently, Table 11 presents a comparison of TMscNet's performance with mainstream algorithms on the Chest X-ray Dataset. Additionally, Fig. 11 illustrates the confusion matrix of TMscNet on Dataset 2, highlighting TMscNet's strong performance on this dataset and further validating its efficacy.

As shown in the metrics comparison in Table 9 and Table 11, TMscNet significantly outperforms the latest models such as Biformer [45] and RMT [44] on both datasets. Biformer is a novel Transformer variant that enhances classification performance by combining local and global features through a dual-line attention mechanism. On the other hand, RMT introduces explicit spatial priors and a decay mechanism in the spatial domain to address the computational complexity and spatial modeling capabilities of Vision Transformers, demonstrating exceptional performance across various vision tasks. TMscNet integrates global information, multi-scale features, and temporal sequence information, making it more efficient in detecting lung lesion areas and providing strong generalization capability. By incorporating TCN, TMscNet excels in capturing long-term dependencies, demonstrating superior performance compared to other temporal models like LSTM and GRU. This advantage is reflected in TMscNet's high performance metrics, indicating its superior ability to accurately identify cases across different datasets.

When analyzing the performance of the TMscNet model, we specifically focused on the relatively lower metrics across different categories in the Dataset 1 and Dataset 2 test sets. For instance, in Dataset 1, the F1-Score for the NORMAL category is 96.57%, which is lower than that for the COVID-19 and PNEUMONIA categories. This suggests that the model may face some misclassification risks when distinguishing normal samples, possibly due to similarities in features between the NORMAL category

and certain pathological states, leading to misclassification. Similarly, in Dataset 2, the Sensitivity for the NORMAL category is 97.62%, significantly lower than that for the other two categories. Further analysis revealed that some normal samples were incorrectly classified as pathological states, indicating that the model performs less effectively in detecting normal samples. These results suggest that while TMscNet performs exceptionally well overall, there is still room for improvement in distinguishing between normal and pathological samples, especially in cases where there is a high potential for confusion.

Although this study used two datasets, which provide valuable diversity for the chest X-ray image classification task, some potential limitations still exist. First, class imbalance is a common issue, especially when there are fewer COVID-19 cases. To address this, the study used data augmentation to balance the number of samples in each category, reducing the impact of class imbalance on model training. Furthermore, image anomalies and label biases in the datasets could also affect the model's training performance. Variations in image quality, different capturing angles, and label errors might lead to inconsistencies in the data. To minimize these impacts, the study carefully examined the data and performed necessary normalization during the training process.

In the experimental design, we assumed that the model's generalization ability would be influenced by the diversity of the dataset. Therefore, to verify this assumption, the study introduced a second dataset for cross-dataset validation to assess the model's performance on different datasets.

Future research could consider expanding the diversity of the datasets by incorporating more samples from different hospitals or regions, and further validating the model's applicability and robustness in various scenarios.

5 Conclusions

Our study addresses a significant challenge presented by the current COVID-19 outbreak: the need for rapid and accurate case screening and diagnosis. Our research focuses on leveraging deep learning techniques, specifically automated detection systems utilizing chest X-ray images, to offer healthcare workers a potent tool for swift identification of potential COVID-19 patients. To achieve this, we introduce TMscNet, a novel deep learning model designed to enhance the classification accuracy of chest X-ray images. TMscNet integrates various techniques, including global response normalization, multi-scale fusion convolutional blocks, and TCN network structure, to better perceive global features, multi-scale information, and temporal information. To conduct qualitative analysis, we employed the Score-CAM

algorithm to visualize the output of TMscNet. The results demonstrate that this method enables more effective utilization of lesion information in chest X-ray images. Through experimental validation on two datasets and thorough analysis of results, we demonstrate TMscNet's significant performance advantages in COVID-19 detection.

Despite acknowledging the strengths of the current model, we also recognize that there is still room for improvement. Future research should focus on further optimizing the model structure, expanding the dataset size, and enhancing the model's interpretability. Moreover, to further improve the model's generalization ability, future work could explore the integration of more diverse datasets and the adoption of advanced data augmentation techniques. The potential of TMscNet is not limited to COVID-19 detection; its strong feature extraction and temporal modeling capabilities can also be extended to other medical image classification tasks, such as lung cancer detection, lung nodule diagnosis, and it may even be applied to the diagnosis of other organs, such as brain tumors and breast cancer.

Author contributions XW conceived of the presented idea, developed the theory, and performed the computations. Additionally, XW authored the manuscript draft. All figures, tables, and experiments were conducted by XW. The other authors (SZ, DW, JZ, HC) contributed to the final version of the manuscript through critical review and editing.

Funding National Natural Science Foundation of China,82074579,National Natural Foundation of China,82374620

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interests The authors declare no competing interests.

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